Lecture 1. Arithmetic, fractions, ratios, and percentages

Notation	Meaning and example
$ab,\ a(b),\ a \cdot b,\ a \times b$	Multiply a by b. For example, $3(4) = 12$. Letters stand for numbers or quantities defined in the problem.
$a/b,\ \dfrac{a}{b},\ a \div b$	Divide a by nonzero b. A fraction bar groups its whole numerator and denominator. For example, $(6 + 2)/4 = 2$.
$(\,),\ [\,]$	Group operations. In $3(8 - 2)$, subtract inside the parentheses before multiplying. A mixed number such as $1\frac{1}{2}$ means $1 + \frac{1}{2}$.
$=,\ \neq,\ \approx$	Equal to; not equal to; approximately equal to. Use $\approx$ when rounding: $1/3 \approx 0.333$.
$<,\ >,\ \leq$	Less than; greater than; less than or equal to. Thus $1 \leq a < 10$ means at least 1 and less than 10.
$a^n,\ 10^{-n}$	Powers: $2^3 = 2 \cdot 2 \cdot 2 = 8$. A negative exponent gives a reciprocal: $10^{-3} = 1/1000$. For $a \neq 0$, $a^0 = 1$.
$a : b,\ p\%$	A ratio; a percentage. A ratio $2 : 3$ has 5 total parts. $p\%$ means $p/100$, so $25\% = 1/4 = 0.25$.
$\longrightarrow$	Changes to the next value. For example, $100 \longrightarrow 120$ shows a value increasing from 100 to 120.
$A_1,\ A_2,\ V_1,\ V_2$	Subscripts label different areas or volumes. They are labels, not powers. A length scale factor is written s.
$P,\ I,\ A,\ r,\ t$	For simple interest: principal, interest earned, final balance, annual rate as a decimal, and time in years. Thus 9 months means $t = 3/4$.
cm^2, cm^3, $\mathrm{kg/m}^3$	Square centimeters, cubic centimeters, and kilograms per cubic meter. A slash in units means per. L, h, s, and mph mean liters, hours, seconds, and miles per hour.
1.3.2; C1	Exercise 2 in Section 1.3; mixed-practice question 1. Solutions use matching labels.

1.1 Fractions and decimals: recognize useful forms

Formula.

$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \cdot \frac{d}{c} \qquad (b, c, d \neq 0).$$

Useful equivalents include

$$\frac{1}{8} = 0.125 = 12.5\%, \quad \frac{1}{4} = 0.25 = 25\%, \quad \frac{3}{8} = 0.375 = 37.5\%, \quad \frac{3}{4} = 0.75 = 75\%.$$

Explanation. Listen for what the question asks: a fraction, a decimal, or a percentage. Choose the form that makes the calculation shortest. A quarter means divide by 4; three-eighths means divide by 8 and multiply by 3. When comparing close fractions, cross-multiply instead of rounding.

For a negative-number question, keep the sign attached to the number. Among negative values, the one closer to zero is greater.

Example. What is 37.5% of 64? Recognize $37.5\% = 3/8$:

$$64 \div 8 \cdot 3 = 24.$$

Harder example. How many three-quarter-cup servings can be poured from six cups? This is a division question:

$$6 \div \frac{3}{4} = 6 \cdot \frac{4}{3} = 8 \text{ servings.}$$

The answer must exceed six because each serving is smaller than one cup.

Practice.

1.1.1. What is 12.5% of 96?

1.1.2. Which is greater: seven-twelfths or five-eighths?

1.1.3. What is one and two-thirds divided by five-sixths?

Solutions.

1.1.1. One-eighth of 96 is 12.

1.1.2. Compare $7 \cdot 8 = 56$ with $5 \cdot 12 = 60$. Five-eighths is greater.

1.1.3. Convert the mixed number: $(5/3)(6/5) = 2$.

1.2 Order of operations and quick products

Formula.

$$a(b + c) = ab + ac, \qquad (a - b)(a + b) = a^2 - b^2.$$

Explanation. Listen carefully for grouping words such as *the quantity* or *the square of*. Within a group, use exponents first, then multiplication and division from left to right, then addition and subtraction from left to right.

For multiplication near a round number, use compensation. For example, multiply by 19 as multiply by 20 and subtract one copy. Use the difference of squares for factors equally far from a convenient center.

Example. What is 19 times 25?

$$20(25) - 25 = 500 - 25 = 475.$$

Harder example. What is 48 times 52? The center is 50:

$$(50 - 2)(50 + 2) = 2500 - 4 = 2496.$$

The shortcut uses one familiar square and one small correction.

Practice.

1.2.1. Evaluate ten minus two times three.

1.2.2. What is 99 times 24?

1.2.3. What is 97 times 103?

Solutions.

1.2.1. Multiply first: $10 - 6 = 4$.

1.2.2. $100(24) - 24 = 2400 - 24 = 2376$.

1.2.3. The center is 100: $100^2 - 3^2 = 9991$.

1.3 Equivalent fractions and cancellation

Formula. For nonzero denominators,

$$\frac{a}{b} = \frac{ac}{bc} \quad (c \neq 0).$$

Divide numerator and denominator by a common factor to reduce a fraction.

Explanation. Reduce before multiplying. This keeps the arithmetic small and makes it easier to remember the intermediate numbers. Cancel only factors in a product, never individual terms in a sum. For addition, find a common denominator; for multiplication, look for cancellation first.

Example. What is three-fourths of 28? Divide before multiplying:

$$28 \div 4 \cdot 3 = 21.$$

Harder example. What is seven-ninths times twenty-seven fourteenths?

$$\frac{7}{9} \cdot \frac{27}{14} = \frac{1}{2} \cdot 3 = \frac{3}{2}.$$

Cancel 7 against 14 and 27 against 9 before multiplying.

Practice.

1.3.1. Express eighteen twenty-fourths in lowest terms.

1.3.2. What is five-sixths times eighteen twenty-fifths?

1.3.3. What is three-fourths plus five-sixths? Give a fraction in lowest terms.

Solutions.

1.3.1. Divide both numbers by 6: $3/4$.

1.3.2. Cancel $18/6 = 3$ and $5/25 = 1/5$. The product is $3/5$.

1.3.3. Use denominator 12: $9/12 + 10/12 = 19/12$.

1.4 Ratios, proportions, and scaling

Formula. A ratio $a : b$ has $a + b$ parts. For similar figures with positive length scale factor s,

$$\frac{A_2}{A_1} = s^2, \qquad \frac{V_2}{V_1} = s^3.$$

Explanation. First identify whether the question gives a total or a difference. A total uses the sum of the ratio parts; a difference uses their difference. The word *per* often signals a unit rate.

For similar figures, listen for whether the requested quantity is length, area, or volume. Length uses the scale factor once, area twice, and volume three times.

Example. Red and blue beads are in the ratio three to five. There are 64 beads. How many are red?

$$64 \div (3 + 5) = 8 \text{ per part}, \qquad 3(8) = 24.$$

Harder example. A rectangle measures 8 by 12 centimeters. Each dimension increases by 25%. What is its new area? Multiply each length by 1.25:

$$8(1.25) = 10, \quad 12(1.25) = 15, \quad A = 150 \text{ cm}^2.$$

Practice.

1.4.1. The ratio of boys to girls is two to three in a group of 35. How many girls are there?

1.4.2. A map uses one centimeter for two kilometers. How many kilometers does 3.5 centimeters represent?

1.4.3. Similar solids have corresponding lengths in the ratio two to three. What is their volume ratio?

Solutions.

1.4.1. Each part is $35/5 = 7$, so there are $3(7) = 21$ girls.

1.4.2. $3.5(2) = 7$ km.

1.4.3. Cube each part: $2^3 : 3^3 = 8 : 27$.

1.5 Percentages and changing bases

Formula. For a rate r written as a decimal,

$$\text{increase multiplier} = 1 + r, \qquad \text{decrease multiplier} = 1 - r.$$

For a positive original amount,

$$\text{percent change} = \frac{\text{change}}{\text{original amount}} \cdot 100\%.$$

Explanation. Use 10%, 5%, 1%, or familiar fractions as building blocks. If helpful, swap the numbers: $a\%$ of b equals $b\%$ of a.

Listen for the base. *What percent of* divides by the stated whole; *percent increase* divides by the original amount. Successive changes use the current value, so multiply their factors.

Example. What is 16% of 25? Swap the numbers: 25% of 16 is 4.

Harder example. A price rises by 20% and then falls by 20%. What is the net percentage change? Choose an original price of 100:

$$100 \longrightarrow 120 \longrightarrow 96.$$

The net result is a 4% decrease.

Practice.

1.5.1. What is 15% of 80?

1.5.2. A quantity increases from 48 to 60. What is the percent increase?

1.5.3. A price falls from \$100 to \$80. What percentage increase restores the original price?

Solutions.

1.5.1. 10% is 8 and 5% is 4, giving 12.

1.5.2. The change is 12; $12/48 = 1/4$, so the increase is 25%.

1.5.3. The increase is 20 on a base of 80: $20/80 = 25\%$.

1.6 Unit conversions and estimation

Formula.

$$1\text{ L} = 1000\text{ cm}^3, \qquad 1\text{ m}^2 = 10{,}000\text{ cm}^2, \qquad 1\text{ km/h} = \frac{5}{18}\text{ m/s}.$$

Explanation. Identify the requested unit before calculating. Change the units, then apply the rate. For area and volume, square or cube the length conversion factor.

Estimate the size before doing exact arithmetic. When asked for the nearest integer, keep enough precision to round at the end. An estimate is a check unless the question requests an approximate answer.

Example. Convert 72 kilometers per hour to meters per second:

$$72 \cdot \frac{5}{18} = 20 \text{ m/s}.$$

Harder example. Water flows at 250 cubic centimeters per second. How many liters flow in two minutes? Two minutes is 120 seconds:

$$250(120) = 30{,}000 \text{ cm}^3 = 30 \text{ L}.$$

Practice.

1.6.1. How many milliliters are in 0.75 liters?

1.6.2. How many square centimeters are in 0.02 square meters?

1.6.3. To the nearest whole number, what is 19.8 times 5.1?

Solutions.

1.6.1. $0.75(1000) = 750$ mL.

1.6.2. $0.02(10{,}000) = 200$ cm^2.

1.6.3. Estimate $20(5) = 100$ first. Exactly, $(20 - 0.2)(5 + 0.1) = 100.98$, which rounds to 101.

1.7 Rates, work, and averages

Formula.

$$\text{average speed} = \frac{\text{total distance}}{\text{total time}}, \qquad \text{average value} = \frac{\text{total of the values}}{\text{number of values}}.$$

Independent workers with constant rates combine by adding the amount each completes per unit time.

Explanation. For a list clustered around a convenient number, average the small deviations from that number. For a group-average problem, convert each average back to a total before combining.

For travel, add the distances and times separately. Equal-distance legs generally do not take equal times, so do not simply average their speeds.

Example. What is the average of 78, 81, and 84? Relative to 81, the deviations are $-3, 0, 3$, which sum to zero. The average is 81.

Harder example. A car travels 60 miles at 30 mph and another 60 miles at 60 mph. What is its average speed?

$$\text{time} = 2 + 1 = 3 \text{ h}, \qquad \text{average speed} = \frac{120}{3} = 40 \text{ mph}.$$

Practice.

1.7.1. A pulse count is 18 beats in 15 seconds. What is the rate in beats per minute?

1.7.2. Five scores average 16. A sixth score is 22. What is the new average?

1.7.3. One machine finishes a job in 6 hours, and another in 3 hours. At constant independent rates, how many hours do they take together?

Solutions.

1.7.1. A minute contains four 15-second intervals, so the rate is $18(4) = 72$ beats/min.

1.7.2. The original total is $5(16) = 80$. The new average is $(80 + 22)/6 = 17$.

1.7.3. In an hour they complete $1/6 + 1/3 = 1/2$ of a job, so they take 2 h.

1.8 Money, discounts, and simple interest

Formula.

$$\text{sale price} = \text{original price}\,(1 - \text{discount rate}), \qquad I = Prt.$$

For simple interest, P is principal, r is the annual rate as a decimal, t is time in years, I is interest, and the final balance is $A = P + I$.

Explanation. For a discount, compute the amount retained or subtract a familiar fraction. If tax follows a discount, apply it to the sale price. For simple interest, find one year's interest and then scale the time. Distinguish interest earned from the final balance.

Example. An \$80 item is 25% off. What is the sale price? Subtract one-quarter of 80:

$$80 - 20 = \$60.$$

Harder example. A \$120 item is discounted 25% and then taxed 10%. What is the final price?

$$120(0.75) = 90, \qquad 90 + 9 = \$99.$$

Practice.

1.8.1. A \$50 item is 20% off. What is its sale price?

1.8.2. How much simple interest does \$600 earn at 5% per year for two years?

1.8.3. What is the final balance on \$800 at 6% annual simple interest after nine months?

Solutions.

1.8.1. Subtract 10: the sale price is \$40.

1.8.2. One year's interest is 30, so two years earn \$60.

1.8.3. Annual interest is 48. Nine months is 3/4 of a year, so interest is 36 and the balance is \$836.

1.9 Simple mixtures and dilution

Formula.

$$\text{solute amount} = \text{concentration as a decimal} \times \text{solution volume}.$$

Add the solute amounts, then divide by the total volume to find the new concentration. Here concentrations are by volume and volumes are assumed additive.

Explanation. With equal volumes, the new concentration is the ordinary average of the two concentrations. With unequal volumes, first find the amount of solute in each portion. Adding pure water leaves the solute amount unchanged.

Listen for whether the question asks for added water, final volume, or concentration. Keep those quantities separate.

Example. Equal volumes of 10% solution and 30% solution are mixed. What is the final concentration? Equal volumes give equal weights:

$$(10\% + 30\%)/2 = 20\%.$$

Harder example. Two liters of 10% solution are mixed with three liters of 20% solution. What is the concentration?

$$\text{solute} = 0.2 + 0.6 = 0.8 \text{ L}, \qquad \frac{0.8}{5} = 16\%.$$

Practice.

1.9.1. How many liters of solute are in five liters of a 20% solution?

1.9.2. Two liters of a 30% solution are mixed with one liter of water. What is the new concentration?

1.9.3. How many liters of water must be added to four liters of a 25% solution to make it 20%?

Solutions.

1.9.1. One-fifth of five liters is 1 L.

1.9.2. Solute stays at 0.6 L. Divide by the new volume, 3 L: the result is 20%.

1.9.3. There is 1 L of solute. At 20%, that requires 5 L total, so add 1 L of water.

1.10 Scientific notation and powers of ten

Formula.

$$(a \times 10^m)(b \times 10^n) = ab \times 10^{m+n}, \qquad \frac{a \times 10^m}{b \times 10^n} = \frac{a}{b} \times 10^{m-n}.$$

In normalized scientific notation, the coefficient has magnitude at least 1 and less than 10.

Explanation. Handle coefficients and exponents separately, then normalize. If the coefficient becomes 20, rewrite it as 2×10^1 and increase the exponent by one.

For addition, express both terms using the same power of ten. Listen carefully to the sign of each exponent; subtracting a negative exponent increases the exponent.

Example. What is four times ten to the third power multiplied by three times ten to the second power?

$$(4 \times 10^3)(3 \times 10^2) = 12 \times 10^5 = 1.2 \times 10^6.$$

Harder example. What is six times ten to the negative fourth power divided by two times ten to the negative sixth power?

$$\frac{6 \times 10^{-4}}{2 \times 10^{-6}} = 3 \times 10^{-4-(-6)} = 3 \times 10^2.$$

Practice.

1.10.1. Write 0.00072 in scientific notation.

1.10.2. Multiply eight times ten to the negative third power by five times ten to the fifth power. Give scientific notation.

1.10.3. Add three times ten to the fifth power and six times ten to the fourth power. Give scientific notation.

Solutions.

1.10.1. 7.2×10^{-4}.

1.10.2. $8(5) = 40$ and $-3 + 5 = 2$, giving $40 \times 10^2 = 4 \times 10^3$.

1.10.3. Rewrite the second term as 0.6×10^5. The sum is 3.6×10^5.

Mixed oral practice

These original practice questions are written for reading aloud. Work without seeing the page, identify the requested quantity, and use the shortest reliable method. The later questions combine two steps; they do not require extra theory.

C1. What is 37.5% of 160?

C2. What is two and one-quarter divided by three-eighths?

C3. What is 49 times 51?

C4. The ratio of red beads to blue beads is five to seven. If there are 96 beads, how many more blue beads than red beads are there?

C5. A price is reduced by 20% to \$72. What was the original price?

C6. A 6-centimeter by 10-centimeter rectangle has both dimensions increased by 50%. What is its new area in square centimeters?

C7. A tank receives water at 300 milliliters per second. How many liters enter in two minutes?

C8. A car travels 120 miles at 40 mph and then 120 miles at 60 mph. What is its average speed in miles per hour?

C9. Four scores average 18. What fifth score makes the average 20?

C10. A \$200 item is discounted 15% and then taxed 10%. What is the final price?

C11. How many liters of water must be added to six liters of a 20% solution to obtain a 15% solution?

C12. Divide eight times ten to the negative third power by two times ten to the negative fifth power. Give the answer in scientific notation.

Mixed-practice solutions

C1. $37.5\% = 3/8$. Divide 160 by 8 and multiply by 3: the answer is 60.

C2. Convert $2\frac{1}{4}$ to $9/4$, then multiply by $8/3$. Cancel before multiplying to get 6.

C3. Use the center 50: $50^2 - 1 = 2499$.

C4. There are 12 parts, each worth $96/12 = 8$. The difference is two parts, or 16 beads.

C5. The sale price is 80% of the original. Since $72/0.8 = 90$, the original price was \$90.

C6. The new lengths are $6(1.5) = 9$ and $10(1.5) = 15$ cm, so the new area is 135 cm^2.

C7. Two minutes is 120 s. The volume is $300(120) = 36{,}000$ mL, or 36 L.

C8. Total time is $120/40 + 120/60 = 3 + 2 = 5$ h. Total distance is 240 miles, so the average is 48 mph.

C9. The first four total 72. Five scores averaging 20 must total 100, so the fifth is $100 - 72 = 28$.

C10. 15% of 200 is 30, so the sale price is 170. Tax adds 17, making the final price \$187.

C11. There is $6(0.20) = 1.2$ L of solute. At 15%, the total volume must be $1.2/0.15 = 8$ L, so add 2 L of water.

C12. Divide coefficients to get 4 and subtract exponents: $-3 - (-5) = 2$. The answer is 4×10^2.